

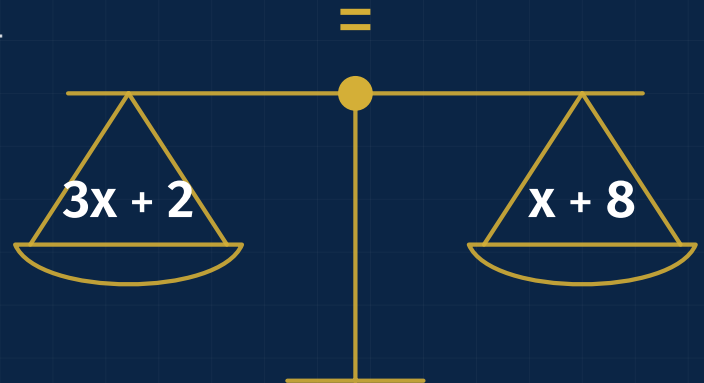


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CHAPTER 2 • ALGEBRA • TOPIC 4

Factorisation

Count the terms, then choose: two terms
→ HCF or DOTS, three terms → the CC
Rule. And “hence solve” is the mark-
earner.



6

VARIANTS

22

PAST-PAPER PARTS

50

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Common factor

V2 DOTS

V3 HCF, then DOTS

V4 Three terms: CC Rule

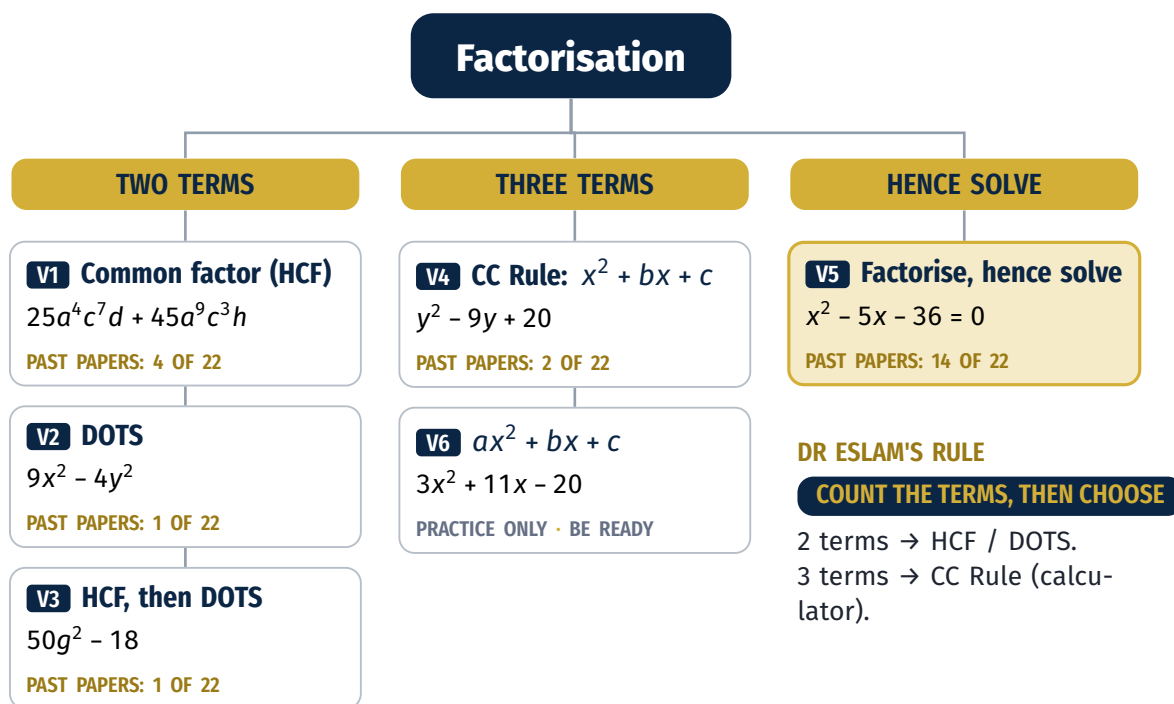
V5 Factorise, hence solve

V6 $ax^2 + bx + c$

TOPIC MAP · EVERY VARIANT ON ONE PAGE

Factorisation at a glance



Your notes


★ How often does each variant come up?

20 past-paper questions (Oct/Nov 2020 → Nov 2025) hold 22 factorising parts; 50 practice parts — grouped by type.

	PAST PAPERS (OUT OF 22)	PRACTICE PARTS
V1 Common factor (HCF)	4	26
V2 DOTS	1	8
V3 HCF, then DOTS	1	3
V4 CC Rule: $x^2 + bx + c$	2	8
V5 Factorise, hence solve	14	1
V6 $ax^2 + bx + c$	0	4

WHAT THIS TELLS YOU “(i) Factorise $x^2 + bx + c$ (ii) Hence solve” is on almost every Paper 1 — 14 of 22 parts, 3 marks. The CC Rule makes it 3 sure marks.

BEFORE YOU START

The big idea



Factorising = expanding backwards.

You are given the answer of a multiplication ($x^2 - 9x + 20$) and must find what was multiplied:

$(x - 4)(x - 5)$.

Always **check by expanding** — you should get the question back.

$$(x - 4)(x - 5) \xrightarrow{\text{EXPAND}} x^2 - 9x + 20$$

$$\xleftarrow{\text{FACTORISE}}$$

Count the terms, then choose

HOW MANY TERMS?

2 TERMS

- Something common? → **HCF** (V1)
- Square – square? → **DOTS** (V2)
- Both? → HCF **first**, then DOTS (V3)

3 TERMS

- $x^2 + bx + c$ → **CC Rule** (V4)
- $ax^2 + bx + c$ → CC Rule + check (V6)
- “Hence solve” → each bracket = 0 (V5)

The CC Rule (Calculator & Check)

1 CALCULATOR

roots of $x^2 - 9x + 20 = 0$

2 FLIP THE SIGNS

roots 4, 5 →
 $(x - 4)(x - 5)$

3 CHECK

first terms make x^2 ?

$$x_1 = 5$$

$$x_2 = 4$$



CALCULATOR FIRST, THEN CHECK

Casio fx-991EX: **MENU** → Equation (**A**) → Polynomial → degree **2**. Enter a, b, c (here 1, -9, 20), press **=** to read the roots. (fx-991ES: **MODE** **5** **3**.)

The roots tell you the brackets; the calculator never tells the examiner — write the brackets.

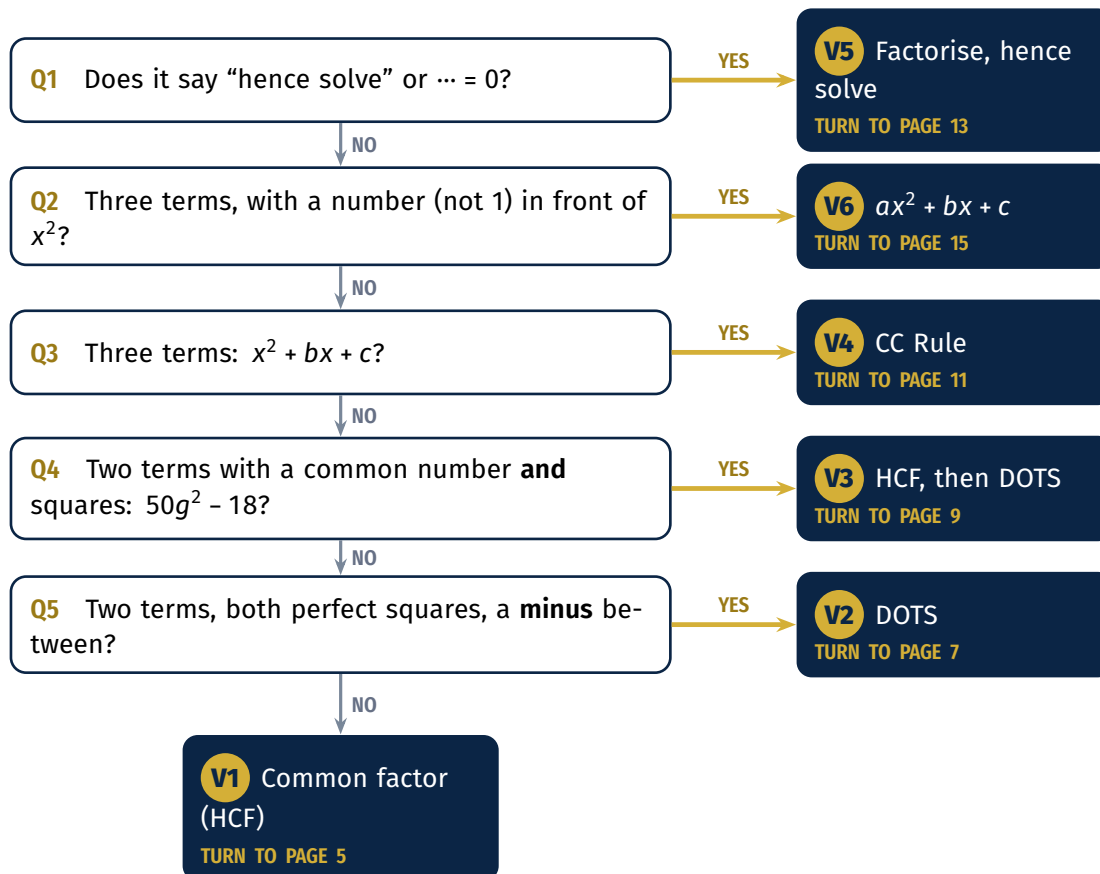
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Count the terms first, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Common Factor (HCF)

Take out the biggest thing every term shares.

SEEN IN PAST PAPERS

4 / 22

PRACTICE QUESTIONS: 26



Your notes

Came up in: Oct/Nov 2020 P1H Q7(b) · May 2021 P1H Q9(a) · May 2023 P1H Q6(c) · Nov 2025 P1H Q10(c)

? What is it?

IN PLAIN WORDS

Find the **Highest Common Factor**: the biggest number that divides every coefficient, and every letter that is in **every** term, at its **lowest power**. Put it outside; what is left goes in the bracket.

IT LOOKS LIKE

$$8p^2 - 2p \quad 12pq - 18p \quad 16a^2b^3 + 20a^3b$$

	number	a	c	d	h	
$25a^4c^7d$	25	a^4	c^7	d	-	
$45a^9c^3h$	45	a^9	c^3	-	h	
HCF	5	a^4	c^3	-	-	$= 5a^4c^3(5c^4d + 9a^5h)$

LOWEST POWER NOT IN BOTH: STAYS INSIDE

Find the HCF column by column: the number, then each letter at its lowest power.

How do I do it?

1 Numbers

The HCF of the numbers: 25 and 45 → 5.

2 Letters

Each letter in **every** term, at its **lowest** power.

3 Fill the bracket

Divide every term by the HCF.

4 Check

Expand back in your head — and nothing common left inside?

BIGGEST NUMBER · LOWEST POWER

“Fully” means the biggest HCF.

V1 Common Factor (HCF)

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Factorise fully” + two terms

→ **Biggest number, lowest power of each letter**

“Explain why Tenzin’s answer is not correct”

→ **Something common is still inside the bracket**“Circle the factor of $x^2 - 5x$ ”→ **Factorise it: $x(x - 5)$, so $x - 5$** **WORKED EXAMPLE** **PAST PAPER**

May 2021 · Paper 1H · Q9(a) · 2 marks

Factorise fully $25a^4c^7d + 45a^9c^3h$ **STEP 1** HCF of 25, 45 = 5

◀ numbers

STEP 2 a^4, c^3 (d, h not in both)

◀ letters: lowest powers

STEP 3 $5a^4c^3(5c^4d + 9a^5h)$ ◀ divide each term by $5a^4c^3$ **STEP 4** expands back to the question; $5c^4d$ and $9a^5h$ share nothing

◀ check

ANSWER $5a^4c^3(5c^4d + 9a^5h)$ ✓ expand: $25a^4c^7d + 45a^9c^3h$ **WATCH OUT**

$\times 2x^2 + 6x = x(2x + 6)$

$\checkmark 2x^2 + 6x = 2x(x + 3)$

The 2 is still common inside $(2x + 6)$ — not fully factorised.**EXAM TIP**

Look inside your bracket: if the terms still share a number or a letter, take it out too.

Check yourself

calculator first · answers on page 17

1.1 Oct/Nov 2020 P1HFactorise fully
 $16m^3g^3 + 24m^2g^5$

Answer:

1.2 Nov 2025 P1HFactorise fully
 $15w^2x^5 + 25w^3x^2$

Answer:

1.3 Practice · EasyTenzin writes
 $2x^2 + 6x = x(2x + 6)$ for
“factorise fully”. Why is
he wrong?

Answer:

V2

VARIANT 2

Difference of Two Squares

Square minus square: $a^2 - b^2 = (a + b)(a - b)$.

SEEN IN PAST PAPERS

1 / 22

PRACTICE QUESTIONS: 8

Your notes

Came up in: Jun 2022 P1H Q11(a)

? What is it?

In plain words

Two terms, a **minus** between, and both are **perfect squares**: $9x^2 = (3x)^2$, $4y^2 = (2y)^2$.

Then $a^2 - b^2 = (a + b)(a - b)$ — one plus bracket, one minus bracket.

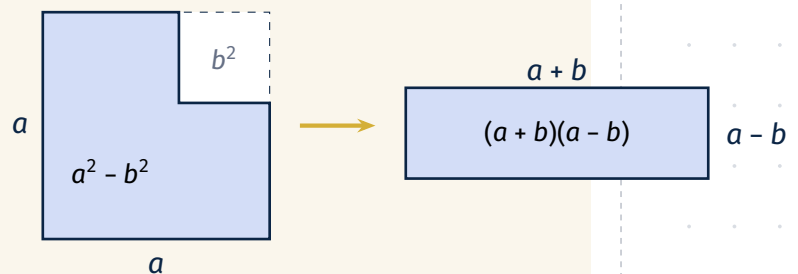
$x^2 + 9$ does **not** factorise: DOTS needs a minus.

IT LOOKS LIKE

$x^2 - 49$

$4c^2 - 9d^2$

$9x^2 - 4y^2$



Cut off the small square, move the strip: same area, now a rectangle.

How do I do it?

1 Check

Two terms? A minus? Both perfect squares? Then it is DOTS.

2 Square roots

Root each term: $\sqrt{9x^2} = 3x$, $\sqrt{4y^2} = 2y$.

3 Write it

$(\text{root} + \text{root})(\text{root} - \text{root})$.

SQUARE - SQUARE Expand to check: the middle terms cancel.

V2 Difference of Two Squares

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DOTwo squares with a minus: $x^2 - 64$ → $(x + 8)(x - 8)$ Letters too: $4c^2 - 9d^2$ → **Root every part:** $(2c + 3d)(2c - 3d)$ “Hence simplify $(x^2 + 4)^2 - (x^2 - 2)^2$ ” → **DOTS with whole brackets as a and b** **WORKED EXAMPLE** **PAST PAPER**

June 2022 · Paper 1H · Q11(a) · 2 marks

Factorise $9x^2 - 4y^2$ **STEP 1** two terms, a minus, $9x^2 = (3x)^2$ and $4y^2 = (2y)^2$ ◀ it is DOTS**STEP 2** $\sqrt{9x^2} = 3x$, $\sqrt{4y^2} = 2y$ ◀ square roots**STEP 3** $(3x + 2y)(3x - 2y)$ ◀ plus bracket, minus bracket**ANSWER** $(3x + 2y)(3x - 2y)$ ✓ expand: $9x^2 - 4y^2$ **WATCH OUT**

✗ $9x^2 - 4y^2 = (3x - 2y)^2$
 ✓ $9x^2 - 4y^2 = (3x + 2y)(3x - 2y)$
 $(3x - 2y)^2$ has a middle term $-12xy$.
 DOTS needs one + and one -.

 EXAM TIP

A 1-mark “circle your answer” DOTS:
 expand each option in your head — only
 $(a + b)(a - b)$ loses the middle.

 Check yourself

calculator first · answers on page 17

2.1 Practice · MediumFactorise $x^2 - 49$

Answer:

2.2 Practice · HardFactorise $4c^2 - 9d^2$

Answer:

2.3 Practice · Hard

Use
 $a^2 - b^2 = (a + b)(a - b)$ to
 simplify
 $(x^2 + 4)^2 - (x^2 - 2)^2$

Answer:

V3

VARIANT 3

HCF, then DOTS

“Factorise fully” two terms: a number comes out first, a DOTS is hiding inside.

SEEN IN PAST PAPERS

1 / 22

PRACTICE QUESTIONS: 3

 Your notes

Came up in: May 2023 P1H Q14

? What is it?**IN PLAIN WORDS**

$50g^2 - 18$ is not two squares — but take out the common **2** and you get $2(25g^2 - 9)$, and **that** is DOTS. “Fully” means: keep going until nothing more factorises. The answer has **three factors**.

IT LOOKS LIKE

$50g^2 - 18$

$20x^2 - 5$

$2e^2 - 18$

$2x^3 - 8x$

$50g^2 - 18$

1 HCF: 2

$2(25g^2 - 9)$

2 DOTS

$2(5g + 3)(5g - 3)$

Two moves, in this order. Each one is a mark.

How do I do it?**1 HCF first**

Take out the common number (and letter):
 $50g^2 - 18 = 2(25g^2 - 9)$.

2 Look inside

Square minus square? $25g^2 = (5g)^2$, $9 = 3^2$.

3 DOTS

$(5g + 3)(5g - 3)$ — keep the 2 in front.

HCF FIRST, THEN DOTS

Three factors in the final answer.

V3 HCF, then DOTS

VARIANT 3 · SEE IT DONE, THEN TRY


 Your notes
 WHEN YOU SEE → YOU DO

"Factorise fully" $50g^2 - 18$ (3 marks) → **HCF, then DOTS: three factors**

A cube: $2x^3 - 8x$ → **HCF $2x$, then DOTS: $2x(x + 2)(x - 2)$**

The numbers are not squares (50, 18) → **Divide by their HCF first — now they are**

 WORKED EXAMPLE **PAST PAPER**

May 2023 · Paper 1H · Q14 · 3 marks

Factorise fully $50g^2 - 18$

STEP 1 $2(25g^2 - 9)$ ◀ HCF of 50 and 18 is 2

STEP 2 $25g^2 = (5g)^2$ and $9 = 3^2$, with a minus ◀ inside is DOTS

STEP 3 $2(5g + 3)(5g - 3)$ ◀ DOTS; keep the 2

ANSWER $2(5g + 3)(5g - 3)$ ✓ expand: $2(25g^2 - 9) = 50g^2 - 18$

 WATCH OUT

✗ $50g^2 - 18 = 2(25g^2 - 9)$ — stop

✓ $2(5g + 3)(5g - 3)$

"Fully": $25g^2 - 9$ still factorises. Stopping early loses the last mark.

 EXAM TIP

Read the marks: 3 marks for two terms means two moves — HCF and DOTS.

 Check yourself

calculator first · answers on page 17

3.1 Practice · HardFactorise fully $20x^2 - 5$

Answer:

3.2 Practice · HardFactorise fully $2e^2 - 18$

Answer:

3.3 Dr EslamFactorise fully $2x^3 - 8x$

Answer:

V4

VARIANT 4

Three Terms: the CC Rule $x^2 + bx + c$: calculator roots, flip the signs, check.

SEEN IN PAST PAPERS

2 / 22

PRACTICE QUESTIONS: 8



Your notes

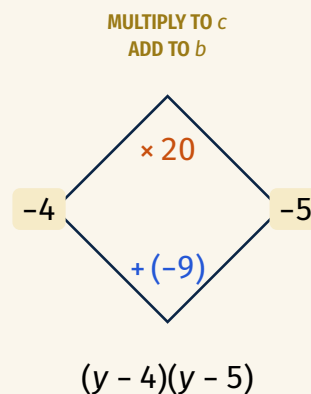
Came up in: Jan 2022 P1HR Q9(c) · May 2023 P1HR Q7(b)

? What is it?**In plain words** $x^2 + bx + c$ factorises into $(x + p)(x + q)$ where $p \times q = c$ and $p + q = b$.Dr Eslam's **CC Rule**: let the **Calculator** find the roots, **flip the signs** to get the brackets, then **Check**.**IT LOOKS LIKE**

$y^2 - 9y + 20$

$x^2 + 3x - 10$

$n^2 - 7n + 12$



No calculator? Find two numbers: product 20, sum -9.

How do I do it?**1 Calculator**Equation mode, degree 2: enter 1, b , c . Read the two roots.**2 Flip the signs**Root 4 $\rightarrow$ bracket $(y - 4)$; root -6 $\rightarrow$ $(y + 6)$.**3 Check**

Expand in your head: the middle term and the number come back?

CC RULE · CALCULATOR & CHECKRoots $r \rightarrow$ brackets $(x - r)$.

V4 Three Terms: the CC Rule

VARIANT 4 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

"Factorise $x^2 + bx + c$ " (2 marks) → **CC Rule: roots → flip the signs**

c positive, b negative: $y^2 - 9y + 20$ → **Both brackets have a minus**

c negative: $x^2 + 3x - 10$ → **One + and one - bracket**

WORKED EXAMPLE **PAST PAPER**

May 2023 · Paper 1HR · Q7(b) · 2 marks

Factorise $y^2 - 9y + 20$

STEP 1 $y^2 - 9y + 20 = 0 \Rightarrow y = 4, y = 5$ ◀ calculator: 1, -9, 20

STEP 2 $(y - 4)(y - 5)$ ◀ flip the signs

STEP 3 $y^2 - 5y - 4y + 20 = y^2 - 9y + 20$ ◀ check by expanding

ANSWER $(y - 4)(y - 5)$ ✓ $(-4) \times (-5) = 20, -4 - 5 = -9$ **WATCH OUT**✗ roots 4, 5 → $(y + 4)(y + 5)$ ✓ roots 4, 5 → $(y - 4)(y - 5)$

Root r makes a bracket $(y - r)$: the sign flips.

**EXAM TIP**

The calculator gives the roots but the exam wants the brackets — write $(y - 4)(y - 5)$, not $y = 4, 5$.

**Check yourself**

calculator first · answers on page 17

4.1

Jan 2022 P1HR

Factorise $n^2 - 7n + 12$

Answer:

4.2

Practice · Medium

Factorise $x^2 + 3x - 10$

Answer:

4.3

Practice · Medium

Factorise $x^2 - 11x + 24$

Answer:

V5**VARIANT 5** MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

14 / 22

PRACTICE QUESTIONS: 1

Factorise, Hence Solve

(i) Factorise, (ii) hence solve = 0: each bracket gives one answer.

Came up in: Oct/Nov 2020 P1H Q7(c) · Jan 2021 P1H Q9(b) · May 2021 P2H Q9 · Nov 2021 P2H Q8(b) · Jan 2022 P1H Q6(b) · Jun 2022 P1HR Q5(b) · Jun 2022 P2H Q11 · May 2023 P1H Q6(d) · May 2024 P1HR Q7(c) · Jun 2024 P2H Q11 · Nov 2024 P1H Q8(a) · Jun 2025 P1HR Q8(c) · Jun 2025 P1H Q6(b) · Nov 2025 P2H Q7


Your notes

? What is it?

In plain words

Part (i) is V4. Part (ii) says **hence solve ... = 0**: if two brackets multiply to 0, **one of them is 0**.

So set **each bracket = 0**. The answers are exactly the calculator roots — but the working must use your brackets.

IT LOOKS LIKE

$$x^2 - 5x - 36 = 0$$

$$y^2 - 11y + 30 = 0$$

$$(x - 9)(x + 4) = 0$$

$$x - 9 = 0$$

$$x + 4 = 0$$

OR

$$x = 9$$

$$x = -4$$

How do I do it?

1 (i) Factorise

CC Rule: calculator roots → brackets (flip the signs).

2 Each bracket = 0

$$(x - 9)(x + 4) = 0 \rightarrow x - 9 = 0 \text{ or } x + 4 = 0.$$

3 Solve both

Two answers: $x = 9$ or $x = -4$ — the same as the calculator.

EACH BRACKET = 0 Two brackets, two answers.

V5 Factorise, Hence Solve

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“(i) Factorise ... (ii) Hence, solve ... = 0” → **Brackets, then each bracket = 0**

“Hence” → **You must use part (i) — not the formula**

3 marks for the whole question → **2 for the brackets, 1 for both answers**

WORKED EXAMPLE **PAST PAPER**

November 2021 · Paper 2H · Q8(b) · 3 marks

(i) Factorise $x^2 - 5x - 36$ (ii) Hence solve $x^2 - 5x - 36 = 0$

STEP 1 $x = 9, x = -4 \Rightarrow (x - 9)(x + 4)$

◀ (i) calculator roots, flip the signs

STEP 2 $(x - 9)(x + 4) = 0$

◀ (ii) use part (i)

STEP 2 $x - 9 = 0$ or $x + 4 = 0$

◀ each bracket = 0

STEP 3 $x = 9$ or $x = -4$

◀ solve both

ANSWER (i) $(x - 9)(x + 4)$ (ii) $x = 9$ or $x = -4$ ✓ $81 - 45 - 36 = 0$ ✓

WATCH OUT

× $(x - 9)(x + 4) = 0 \Rightarrow x = -9, x = 4$

✓ $x = 9$ or $x = -4$

The signs flip back: bracket $(x - 9)$ gives $x = +9$.

EXAM TIP

14 of the 22 past-paper parts! Your solutions must equal the calculator's roots — if not, a sign is wrong.

Check yourself

calculator first · answers on page 17

5.1 Jan 2021 P1H

Factorise $x^2 + 5x - 36$.
Hence solve
 $x^2 + 5x - 36 = 0$.

Answer:

5.2 Jun 2025 P1H

Factorise $y^2 - 11y + 30$.
Hence solve
 $y^2 - 11y + 30 = 0$.

Answer:

5.3 May 2024 P1HR

Factorise $x^2 - 2x - 63$.
Hence solve
 $x^2 - 2x - 63 = 0$.

Answer:

V6

VARIANT 6

Harder: $ax^2 + bx + c$

A number in front of x^2 : a fraction root gives a bracket like $(3x - 4)$.

SEEN IN PAST PAPERS

0 / 22

NOT ASKED YET · BE READY
PRACTICE QUESTIONS: 4


Your notes
? What is it?**In plain words**

With $3x^2$ at the front, one bracket starts with $3x$. The CC Rule still works: a **fraction root** $\frac{a}{b}$ gives the bracket $(bx - a)$.

Then **check**: the first terms must multiply to $3x^2$.

IT LOOKS LIKE

$3x^2 + 11x - 20$

$2p^2 - p - 10$

$6y^2 - y - 5$

$$\begin{array}{l} x_1 = \frac{4}{3} \\ x_2 = -5 \end{array}$$

$(3x - 4)$

$(x + 5)$

ROOT $\frac{4}{3}$: BOTTOM 3 GOES WITH x ,
TOP 4 GETS THE FLIPPED SIGN

How do I do it?**1 Calculator**Equation mode: a , b , c . Read the roots.**2 Fraction root** $\frac{4}{3} \rightarrow (3x - 4)$. Whole root $-5 \rightarrow (x + 5)$.**3 Check the front** $3x \times x = 3x^2$ ✓. If short, a number goes in front.**4 Check the middle**Expand: $3x^2 + 15x - 4x - 20 = 3x^2 + 11x - 20$.

ROOT $a/b \rightarrow (bx - a)$ Then always check the first terms.

V6 Harder: $ax^2 + bx + c$

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DOA number in front of x^2 : $3x^2 + \dots$ → **CC Rule, then check the first terms**A fraction root like $\frac{5}{2}$ → **Bracket** $(2x - 5)$ Two whole roots but $a \neq 1$: $4x^2 - 10x + 6$ → **HCF first:** $2(2x^2 - 5x + 3)$ **WORKED EXAMPLE****PRACTICE**

Factorisation · Hard Q11 · 2 marks

Factorise $3x^2 + 11x - 20$ **STEP 1** $x = \frac{4}{3}, x = -5$

◀ calculator: 3, 11, -20

STEP 2 $(3x - 4)(x + 5)$ ◀ root $\frac{4}{3} \rightarrow (3x - 4)$; root $-5 \rightarrow (x + 5)$ **STEP 3** $3x \times x = 3x^2 \checkmark$

◀ check the first terms

STEP 4 $3x^2 + 15x - 4x - 20 = 3x^2 + 11x - 20$

◀ check the middle

ANSWER $(3x - 4)(x + 5)$

✓ expands back to the question

**WATCH OUT**× roots $\frac{4}{3}, -5 \rightarrow (x - \frac{4}{3})(x + 5)$ ✓ $(3x - 4)(x + 5)$ $(x - \frac{4}{3})(x + 5)$ only gives x^2 at the front — multiply the bracket by 3.**EXAM TIP**

Not on the past papers yet, but in the practice sets — and it comes back in Topic 6 (algebraic fractions).

**Check yourself**

calculator first · answers on page 17

6.1

Practice · Hard

Factorise $3x^2 + 14x + 8$

Answer:

6.2

Practice · Hard

Factorise $2p^2 - p - 10$

Answer:

6.3

Practice · Hard

Factorise $6y^2 - y - 5$

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

1.1 $8m^2g^3(2m + 3g^2)$

1.2 $5w^2x^2(3x^3 + 5w)$

1.3 2 is still common:
 $2x(x + 3)$

2.1 $(x + 7)(x - 7)$

2.2 $(2c + 3d)(2c - 3d)$

2.3
 $(2x^2 + 2) \times 6 = 12x^2 + 12$

3.1 $5(2x + 1)(2x - 1)$

3.2 $2(e + 3)(e - 3)$

3.3 $2x(x + 2)(x - 2)$

4.1 $(n - 3)(n - 4)$

4.2 $(x + 5)(x - 2)$

4.3 $(x - 3)(x - 8)$

5.1 $(x + 9)(x - 4)$; $x = -9$
or 4

5.2 $(y - 5)(y - 6)$; $y = 5$
or 6

5.3 $(x - 9)(x + 7)$; $x = 9$
or -7

6.1 $(3x + 2)(x + 4)$

6.2 $(2p - 5)(p + 2)$

6.3 $(6y + 5)(y - 1)$

THE WHOLE TOPIC ON ONE PAGE

Factorisation cheat sheet



Your notes
V1 Common factor 4/22

$$25a^4c^7d + 45a^9c^3h = 5a^4c^3(\dots)$$

Biggest number, lowest power of each letter.

V2 DOTS 1/22

$$9x^2 - 4y^2 = (3x + 2y)(3x - 2y)$$

Square – square. Needs a minus.

V3 HCF, then DOTS 1/22

$$50g^2 - 18 = 2(5g + 3)(5g - 3)$$

Number out first, then DOTS. Three factors.

V4 CC Rule 2/22

$$y^2 - 9y + 20 = (y - 4)(y - 5)$$

Calculator roots → flip the signs → check.

V5 Hence solve 14/22

$$(x - 9)(x + 4) = 0 \Rightarrow x = 9, -4$$

Each bracket = 0.

V6 $ax^2 + bx + c$

$$3x^2 + 11x - 20 = (3x - 4)(x + 5)$$

Root $\frac{a}{b} \rightarrow (bx - a)$; check the first terms.

**TOP 3 LOST MARKS**

1. $x(2x + 6)$ — not fully factorised ×
2. roots 4, 5 → $(x + 4)(x + 5) \times$
3. $(x - 9)(x + 4) = 0 \Rightarrow x = -9, 4 \times$

**CALCULATOR FIRST, THEN CHECK**

MENU Equation → Polynomial → **2**: enter $a, b, c \rightarrow$ roots.
Root $r \rightarrow (x - r)$; root $\frac{a}{b} \rightarrow (bx - a)$. Expand to check.

BOTTOM LINE Count the terms → 2: HCF / DOTS · 3: CC Rule · “hence solve”: each bracket = 0.